VLAN

show vlan brief, show vlan summary, show interfaces fa0/18 switchport

Normal Range ID: 1 – 1005 (1002-1005 reserved)

Extended Range ID: 1006 – 4096

Založení VLAN:

S3(config)# vlan 20

S3(config-vlan)# name student

S3(config-vlan)# vlan 150

S3(config-vlan)# name VOICE

Přiřazení portu do VLAN:

S3(config)# interface fa0/18

S3(config-if)# switchport mode access

S3(config-if)# switchport access vlan 20

S3(config-if)# mls qos trust cos

S3(config-if)# switchport voice vlan 150

Vyhození portu z VLAN:

S1(config)# interface fa0/18

S1(config-if)# no switchport access vlan

S1(config-if)# end

Vymazání SWICH:

erase startup-config

delete vlan.dat

Nastavení portu jako TRUNK:

S1(config)# interface fastEthernet 0/1

S1(config-if)# switchport mode trunk

S1(config-if)# switchport trunk native vlan 99

S1(config-if)# switchport trunk allowed vlan 10,20,30,99

S1(config-if)# end

Vyhození portu z TRUNK a NATIVE VLAN:

S1(config)# interface fa0/1

S1(config-if)# no switchport trunk allowed vlan

S1(config-if)# no switchport trunk native vlan

S1(config-if)# end

DTP:

S1(config-if)# switchport mode trunk

S1(config-if)# switchport nonegotiate

Inter-VLAN Routing: Router-on-a-Stick

1/ Create and name the VLANs

S1(config)# vlan 99

S1(config-vlan)# name Management

S1(config-vlan)# exit

2/ Create the management interface

S1(config)# interface vlan 99

S1(config-if)# ip add 192.168.99.2 255.255.255.0

S1(config-if)# no shut

S1(config-if)# exit

S1(config)# ip default-gateway 192.168.99.1

3/ Configure access ports

S1(config)# interface fa0/6

S1(config-if)# switchport mode access

S1(config-if)# switchport access vlan 10

S1(config-if)# no shut

4/ Configure trunking ports

S1(config)# interface fa0/1

S1(config-if)# switchport mode trunk

S1(config-if)# no shut

S1(config-if)# exit

R1(config)# interface G0/0/1.10

R1(config-subif)# description Default Gateway for VLAN 10

R1(config-subif)# encapsulation dot1Q 10

R1(config-subif)# ip add 192.168.10.1 255.255.255.0

R1(config-subif)# exit

R1(config)# interface G0/0/1.99

R1(config-subif)# description Default Gateway for VLAN 99

R1(config-subif)# encapsulation dot1Q 99

R1(config-subif)# ip add 192.168.99.1 255.255.255.0

R1(config-subif)# exit

R1(config)# interface G0/0/1

R1(config-if)# description Trunk link to S1

R1(config-if)# no shut

R1(config-if)# end

show interfaces trunk

show interfaces

show ip interface brief

Inter-VLAN Routing: Layer 3 Switch

1/ Create the VLANs

MLS(config)# vlan 10

MLS(config-vlan)# name LAN10

2/ Create the SVI VLAN interfaces

MLS(config)# interface vlan 10

MLS(config-if)# description Default Gateway SVI for 192.168.10.0/24

MLS(config-if)# ip add 192.168.10.1 255.255.255.0

MLS(config-if)# no shut

MLS(config-if)# exit

3/ Configure access ports

MLS(config)# interface GigabitEthernet1/0/6

MLS(config-if)# description Access port to PC1

MLS(config-if)# switchport mode access

MLS(config-if)# switchport access vlan 10

MLS(config-if)# exit

4/ Enable IP routing

MLS(config)# ip routing

5/ Configure the routed port

MLS(config)# interface GigabitEthernet0/0/1

MLS(config-if)# description routed Port Link to R1

MLS(config-if)# no switchport

MLS(config-if)# ip address 10.10.10.2 255.255.255.0

MLS(config-if)# no shut

EtherChannel

PAgP – on, desirable, auto

LACP - on, active, passive

S1(config)# **interface range FastEthernet 0/1 - 2**

S1(config-if-range)# **channel-group 1 mode active**

Creating a port-channel interface Port-channel 1

S1(config-if-range)# **exit**

S1(config-if)# **interface port-channel 1**

S1(config-if)# **switchport mode trunk**

S1(config-if)# **switchport trunk allowed vlan 1,2,20**

Verify

S1# show interfaces port-channel 1

S1# show etherchannel summary

S1# show etherchannel port-channel

S1# show interfaces f0/1 etherchannel

Rozbití EtherChannel

S1(config)# **no interface port-channel 1**

DHCPv4 server on R1

R1(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.9

R1(config)# ip dhcp pool LAN-POOL-1

R1(dhcp-config)# network 192.168.10.0 255.255.255.0

R1(dhcp-config)# default-router 192.168.10.1

R1(dhcp-config)# dns-server 192.168.11.5

R1(dhcp-config)# domain-name example.com

R1(dhcp-config)# end

Verify

R1# show run | section dhcp

R1# show ip dhcp binding

R1# show ip dhcp server statistics

C:> ipconfig /all

Disable/Enable DHCP

R1(config)# no service dhcp

R1(config)# service dhcp

Přehled značení topologií a VLAN schémat

Standardizované profily značení prvků, VLAN a adresních rozsahů v cvičeních:

Název v topologii (Router / MLS / Switch):

0 – RTR-1, MSW-1, SW-1, SW-2, SW-3 (PC1–PC9, PC99)

1 – R-1, MULTI-1, S-1, S-2, S-3 (Host 1–9, Admin Host)

2 – EdgeRouter, L3-SW1, Sw-A, Sw-B, Sw-C (WS-1.1–WS-2.3, Mgmt PC)

Název VLAN 10 v tabulce VLAN:

0 – FL1, FL2, FL3, BDG4, BDG5, BDG6, Management (VLAN 99)

1 – SALES, ACCT, EXEC, FAC, FAB1, FAB2, ADMIN (VLAN 99)

2 – B1F1, B1F2, B1F4, B3, B4, B5, NetAdmin (VLAN 99)

Rozsah IP adres v tabulce adresování:

0 – 192.168.0.0 / 16 (LAN gateway .1, SVI /24, /28)

1 – 172.31.0.0 / 16 (LAN gateway .1, SVI /24, /28)

2 – 10.0.0.0 / 16 (LAN gateway .1, SVI /24, /28)

Příkazy pro ověření konfigurace prvků a VLAN:

Switch# show vlan brief

Switch# show interfaces trunk

Switch# show etherchannel summary

Router# show ip interface brief

Router# show ip route

Konfigurace výchozí brány na koncových stanicích:

Desktop -> IP Configuration -> Default Gateway odpovídající příslušné VLAN

DHCPv4 Relay

R1(config)# interface g0/0/0

R1(config-if)# ip helper-address 192.168.11.6

R1(config-if)# end

Verify

C:> ipconfig /release

C:> ipconfig /renew

C:> ipconfig /all

R1# show ip interface g0/0/0

DHCP client on R2

R2(config)# interface G0/0/1

R2(config-if)# ip address dhcp

R2(config-if)# no shutdown

DHCPv6 Statelless Server on R1

R1(config)# ipv6 unicast-routing

R1(config)# ipv6 dhcp pool IPV6-STATELESS

R1(config-dhcpv6)# dns-server 2001:db8:acad:1::254

R1(config-dhcpv6)# domain-name example.com

R1(config-dhcpv6)# exit

R1(config)# interface GigabitEthernet0/0/1

R1(config-if)# description Link to LAN

R1(config-if)# ipv6 address fe80::1 link-local

R1(config-if)# ipv6 address 2001:db8:acad:1::1/64

R1(config-if)# ipv6 nd other-config-flag

R1(config-if)# ipv6 dhcp server IPV6-STATELESS

R1(config-if)# no shut

R1(config-if)# end

Configure a Stateless DHCPv6 Client

R3(config)# ipv6 unicast-routing

R3(config)# interface g0/0/1

R3(config-if)# ipv6 enable

R3(config-if)# ipv6 address autoconfig

R3(config-if)# end

R3# show ipv6 interface brief

R3# show ipv6 dhcp interface g0/0/1

Configure a Stateful DHCPv6 Server

R1(config)# ipv6 unicast-routing

R1(config)# ipv6 dhcp pool IPV6-STATEFUL

R1(config-dhcpv6)# address prefix 2001:db8:acad:1::/64

R1(config-dhcpv6)# dns-server 2001:4860:4860::8888

R1(config-dhcpv6)# domain-name example.com

R1(config-dhcpv6)#

R1(config)# interface GigabitEthernet0/0/1

R1(config-if)# description Link to LAN

R1(config-if)# ipv6 address fe80::1 link-local

R1(config-if)# ipv6 address 2001:db8:acad:1::1/64

R1(config-if)# ipv6 nd managed-config-flag

R1(config-if)# ipv6 nd prefix default no-autoconfig

R1(config-if)# ipv6 dhcp server IPV6-STATEFUL

R1(config-if)# no shut

R1(config-if)# end

Configure a Stateful DHCPv6 Client

R3(config)# ipv6 unicast-routing

R3(config)# interface g0/0/1

R3(config-if)# ipv6 enable

R3(config-if)# ipv6 address dhcp

R3(config-if)# end

R3# show ipv6 interface brief

R3# show ipv6 dhcp interface g0/0/1

DHCPv6 Relay Agent

R1(config)# interface gigabitethernet 0/0/1

R1(config-if)# ipv6 dhcp relay destination 2001:db8:acad:1::2 G0/0/0

R1(config-if)# exit

Port Security

Secure Unused Ports

S1(config)# interface range fa0/8 - 24

S1(config-if-range)# shutdown

Enable Port Security

S1(config)# interface f0/1

S1(config-if)# switchport port-security

Command rejected: FastEthernet0/1 is a dynamic port.

S1(config-if)# switchport mode access

S1(config-if)# switchport port-security

S1(config-if)# end

Limit and Learn MAC Addresses

S1(config)# interface fa0/1

S1(config-if)# switchport mode access

S1(config-if)# switchport port-security

S1(config-if)# switchport port-security maximum 4

S1(config-if)# switchport port-security mac-address aaaa.bbbb.1234

S1(config-if)# switchport port-security mac-address sticky

S1(config-if)# end

Port Security Aging

S1(config)# interface fa0/1

S1(config-if)# switchport port-security aging time 10

S1(config-if)# switchport port-security aging type **inactivity**

S1(config-if)# end

Pozn.

Absolute – The secure addresses on the port are deleted after the specified aging time.

Inactivity – The secure addresses on the port are deleted only if they are inactive for the specified aging time.

Port Security Violation Modes

S1(config)# interface f0/1

S1(config-if)# switchport port-security violation **restrict**

Security Violation Mode Comparison

Violation Mode	Discards Offending Traffic	Sends Syslog Message	Increase Violation Counter	Shuts Down Port
Protect	Yes	No	No	No
Restrict	Yes	Yes	Yes	No
Shutdown	Yes	Yes	Yes	Yes

Ports in error-disabled State

S1(config)# **interface fa0/18**

S1(config-if)# **shutdown**

S1(config-if)# **no shutdown**

Verify Port Security

S1# show port-security

S1# show port-security interface f0/1

S1# show port-security address

Steps to Mitigate VLAN Hopping Attacks

Step 1: Disable DTP (auto trunking) negotiations on non-trunking ports by using the **switchport mode access** interface configuration command.

Step 2: Disable unused ports and put them in an unused VLAN.

Step 3: Manually enable the trunk link on a trunking port by using the **switchport mode trunk** command.

Step 4: Disable DTP (auto trunking) negotiations on trunking ports by using the **switchport nonegotiate** command.

Step 5: Set the native VLAN to a VLAN other than VLAN 1 by using the **switchport trunk native vlan** vlan_number command.

For example, assume the following:

- FastEthernet ports 0/1 through fa0/16 are active access ports
- FastEthernet ports 0/17 through 0/24 are not currently in use
- FastEthernet ports 0/21 through 0/24 are trunk ports.

S1(config)# interface range fa0/1 - 16

S1(config-if-range)# switchport mode access

S1(config-if-range)# exit

S1(config)# interface range fa0/17 - 20

S1(config-if-range)# switchport mode access

S1(config-if-range)# switchport access vlan 1000

S1(config-if-range)# exit

S1(config)# interface range fa0/21 - 24

S1(config-if-range)# switchport mode trunk

S1(config-if-range)# switchport nonegotiate

S1(config-if-range)# switchport trunk native vlan 999

S1(config-if-range)# end

Mitigate DHCP Attacks

S1(config)# ip dhcp snooping

S1(config)# interface f0/1

S1(config-if)# ip dhcp snooping trust

S1(config-if)# exit

S1(config)# interface range f0/5 - 24

S1(config-if-range)# ip dhcp snooping limit rate 6

S1(config-if)# exit

S1(config)# ip dhcp snooping vlan 5,10,50-52

S1(config)# end

S1# **show ip dhcp snooping**

Mitigate ARP Attacks

S1(config)# ip dhcp snooping

S1(config)# ip dhcp snooping vlan 10

S1(config)# ip arp inspection vlan 10

S1(config)# interface fa0/24

S1(config-if)# ip dhcp snooping trust

S1(config-if)# ip arp inspection trust

S1(config)# ip arp inspection validate src-mac

S1(config)# ip arp inspection validate dst-mac

S1(config)# ip arp inspection validate ip

S1(config)# do show run | include validate

ip arp inspection validate ip

S1(config)# ip arp inspection validate src-mac dst-mac ip

S1(config)# do show run | include validate

ip arp inspection validate src-mac dst-mac ip

Mitigate STP Attacks

Configure PortFast

S1(config)# interface fa0/1

S1(config-if)# switchport mode access

S1(config-if)# spanning-tree portfast

S1(config-if)# exit

S1(config)# spanning-tree portfast default

S1(config)# exit

Configure BPDU Guard

S1(config)# interface fa0/1

S1(config-if)# spanning-tree bpduguard enable

S1(config-if)# exit

S1(config)# spanning-tree portfast bpduguard default

S1(config)# end

S1# show spanning-tree summary

Static Routing

Router(config)# ip route network-address subnet-mask { ip-address | exit-intf [ip-address]} [distance]

Router(config)# ipv6 route ipv6-prefix/prefix-length {ipv6-address | exit-intf [ipv6-address]} [distance]

IPv4 Fully Specified Static Route

ip route 172.16.1.0 255.255.255.0 GigabitEthernet 0/0/1 172.16.2.2

IPv6 Fully Specified Static Route

R1(config)# ipv6 route 2001:db8:acad:1::/64 s0/1/0 fe80::2

Pozn.: Interface has to be specified for a link-local nexthop fe80::2

IPv4 Directly Connected Static Route

R1(config)# ip route 172.16.1.0 255.255.255.0 s0/1/0

IPv6 Directly Connected Static Route

R1(config)# ipv6 route 2001:db8:acad:1::/64 s0/1/0

IPv4 Default Static Route

Router(config)# ip route 0.0.0.0 0.0.0.0 {ip-address | exit-intf}

IPv6 Default Static Route

Router(config)# ipv6 route ::/0 {ipv6-address | exit-intf}

Verify a Static Route

show ip route static

show ip route network

show running-config | section ip route

Floating Static Routes

R1(config)# ip route 0.0.0.0 0.0.0.0 10.10.10.2 **5**

R1(config)# ipv6 route ::/0 2001:db8:feed:10::2 **5**

Administrative Distance

Route Source	Administrative Distance
Directly connected	0
Static route	1
EIGRP summary route	5
External BGP	20
Internal EIGRP	90
OSPF	110
IS-IS	115
RIP	120
External EIGRP	170
Internal BGP	200

Configure Static Host Routes

Branch(config)# ip route 209.165.200.238 **255.255.255.255** 198.51.100.2

Branch(config)# ipv6 route 2001:db8:acad:2::238/**128** 2001:db8:acad:1::2

Branch(config)# exit